

Motor Control Board

Variant: DRAFT

2026-09-04

Rev + (Unreleased)

Page	Index	Page	Index	Page	Index	Page	Index
1	Cover Page	11	Motor Control 4	21	Misc - Board Version, Switch	31	
2	Block Diagram	12	Motor Control 5	22	Power - Sequencing	32	
3	Project Architecture	13	Motor Control 6	23	Revision History	33	
4	Power - Connectors	14	Interface - USB 2.0	24		34	
5	Power - Generation	15	Interface - FD-CAN	25		35	
6	MCU - Power	16	Interface - Interconnects	26		36	
7	MCU - IOs	17	User - LED Indicators & Buzzer	27		37	
8	Motor Control 1	18	Sensing - Temperature & Humidity	28		38	
9	Motor Control 2	19	Sensing - Battery	29		39	
10	Motor Control 3	20	Misc - Holes, Fiducials	30		40	

TOP VIEW

kibot_Mimage_png_3d_viewer_angled_top

BOTTOM VIEW

kibot_Mimage_png_3d_viewer_angled_bottom

NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Block Diagram

File: Block Diagram.kicad_sch

Revision History

File: Revision History.kicad_sch

Project Architecture

File: Project Architecture.kicad_sch

Power - Sequencing

File: Power - Sequencing.kicad_sch

DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for
informational design
notes.

DESIGN NOTE:
Example text for
debug notes.

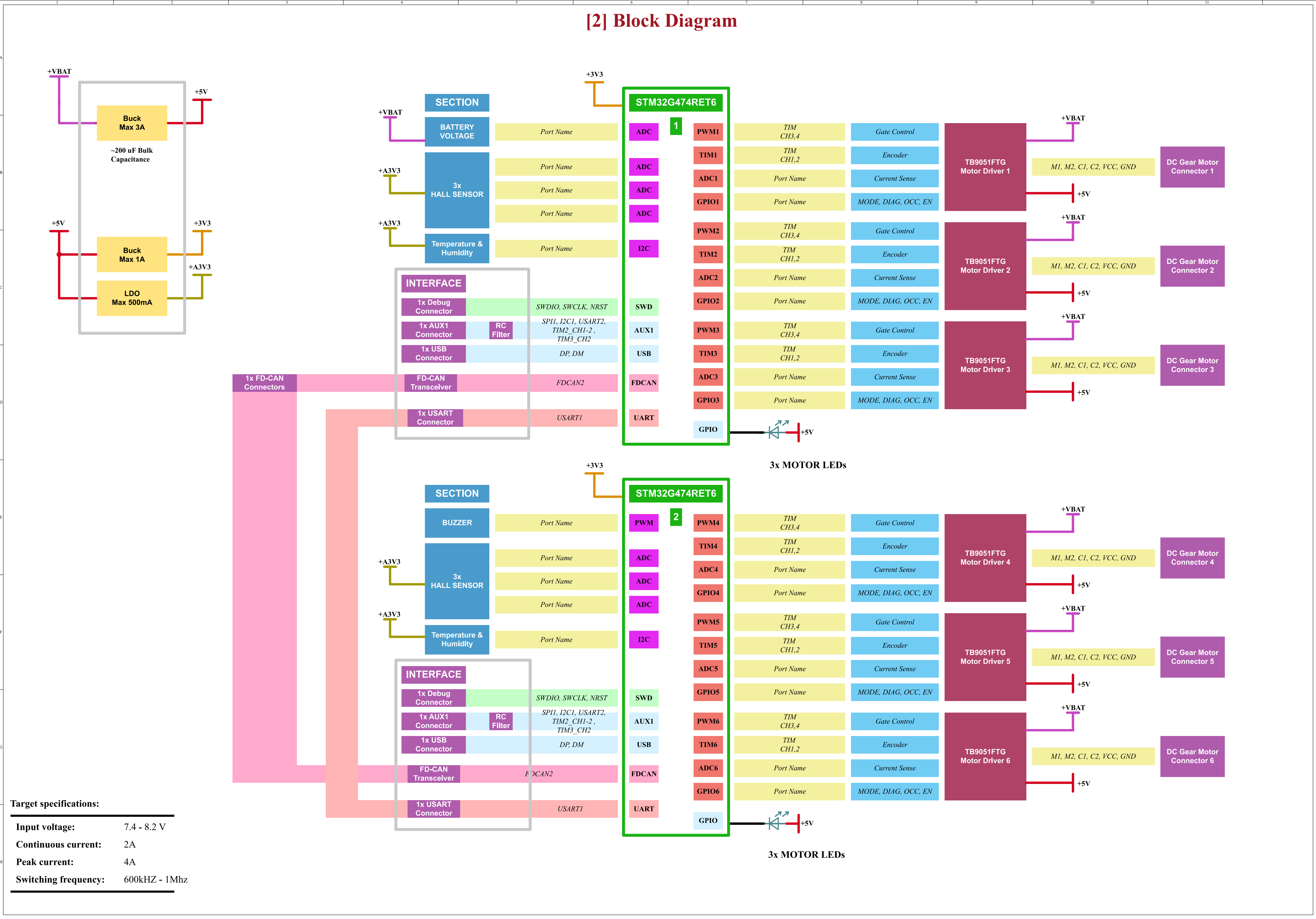
DESIGN NOTE:
Example text for
cautionary design
notes.

DESIGN NOTE:
Example text for
critical design
notes.

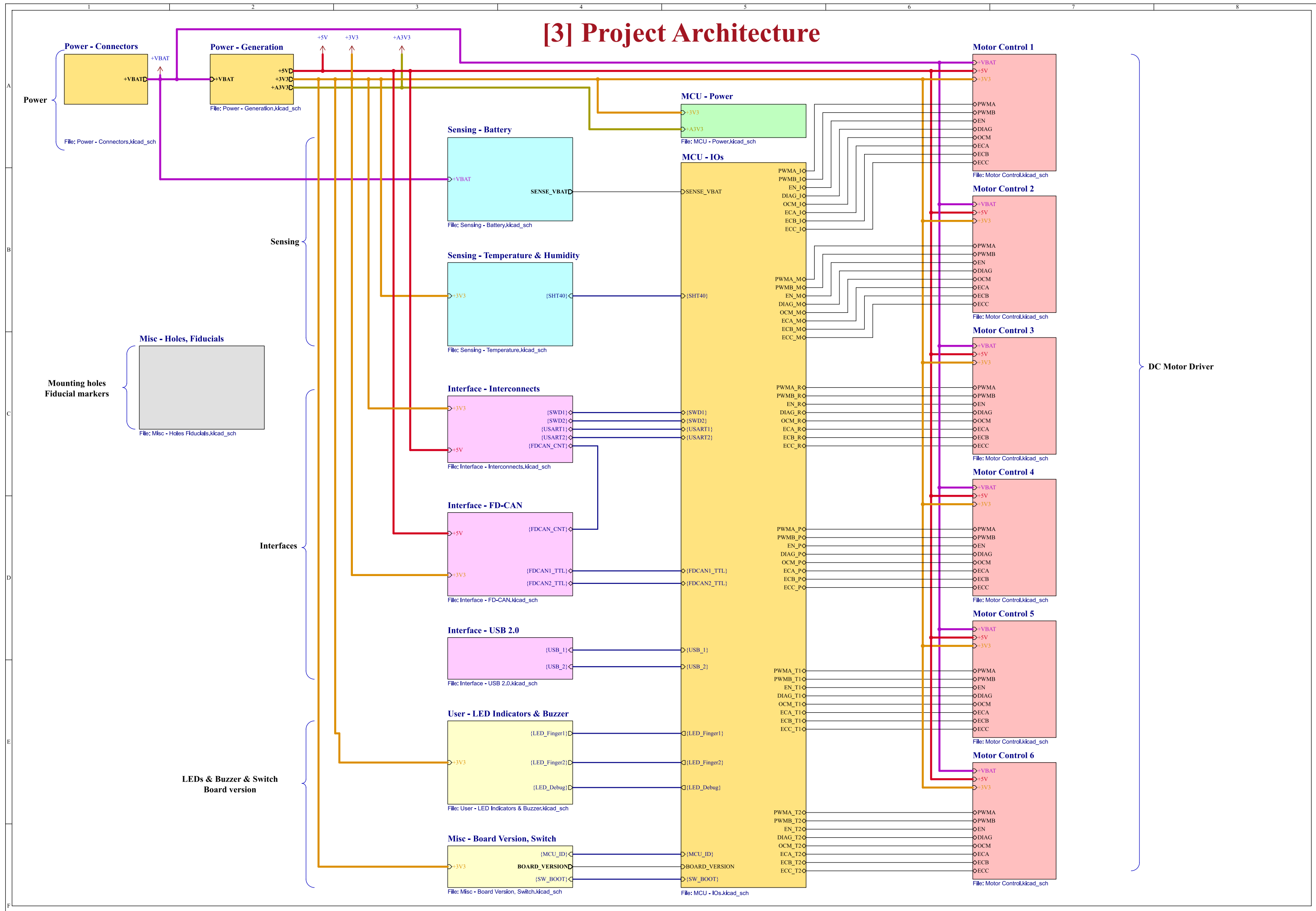
LAYOUT NOTE:
Example text for
critical layout
guidelines.

Date: 04-Sep-2026

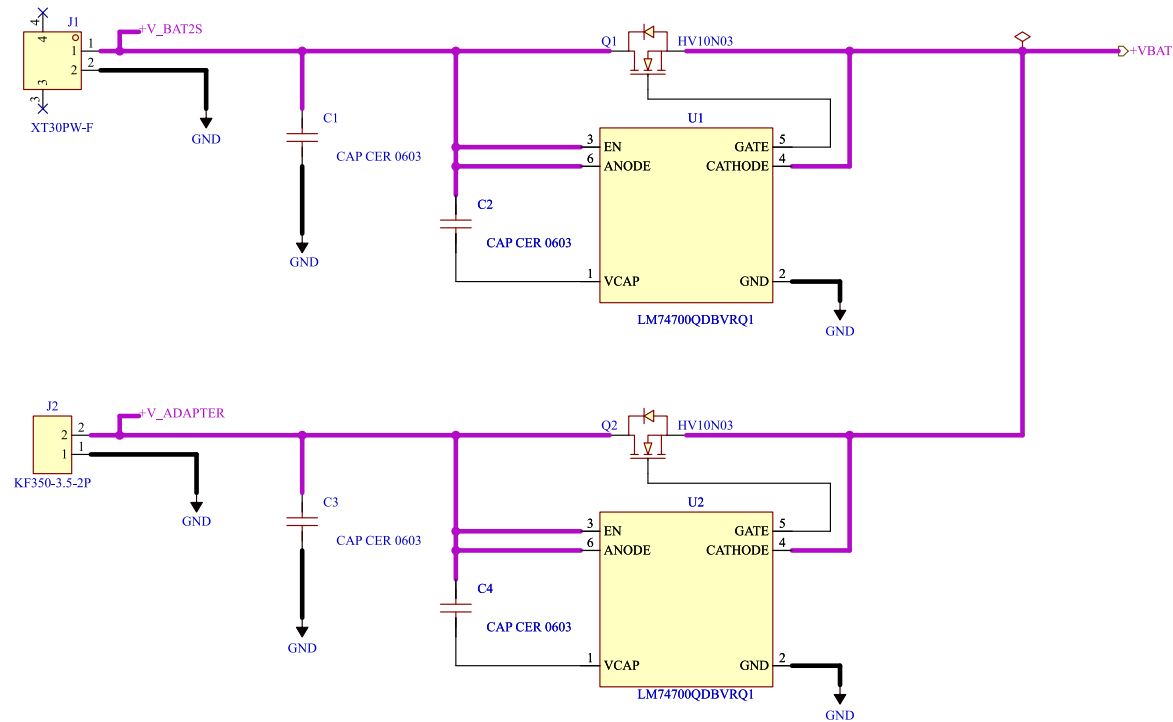
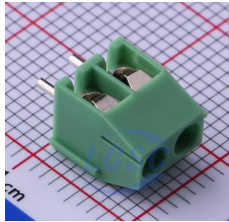
[2] Block Diagram



[3] Project Architecture

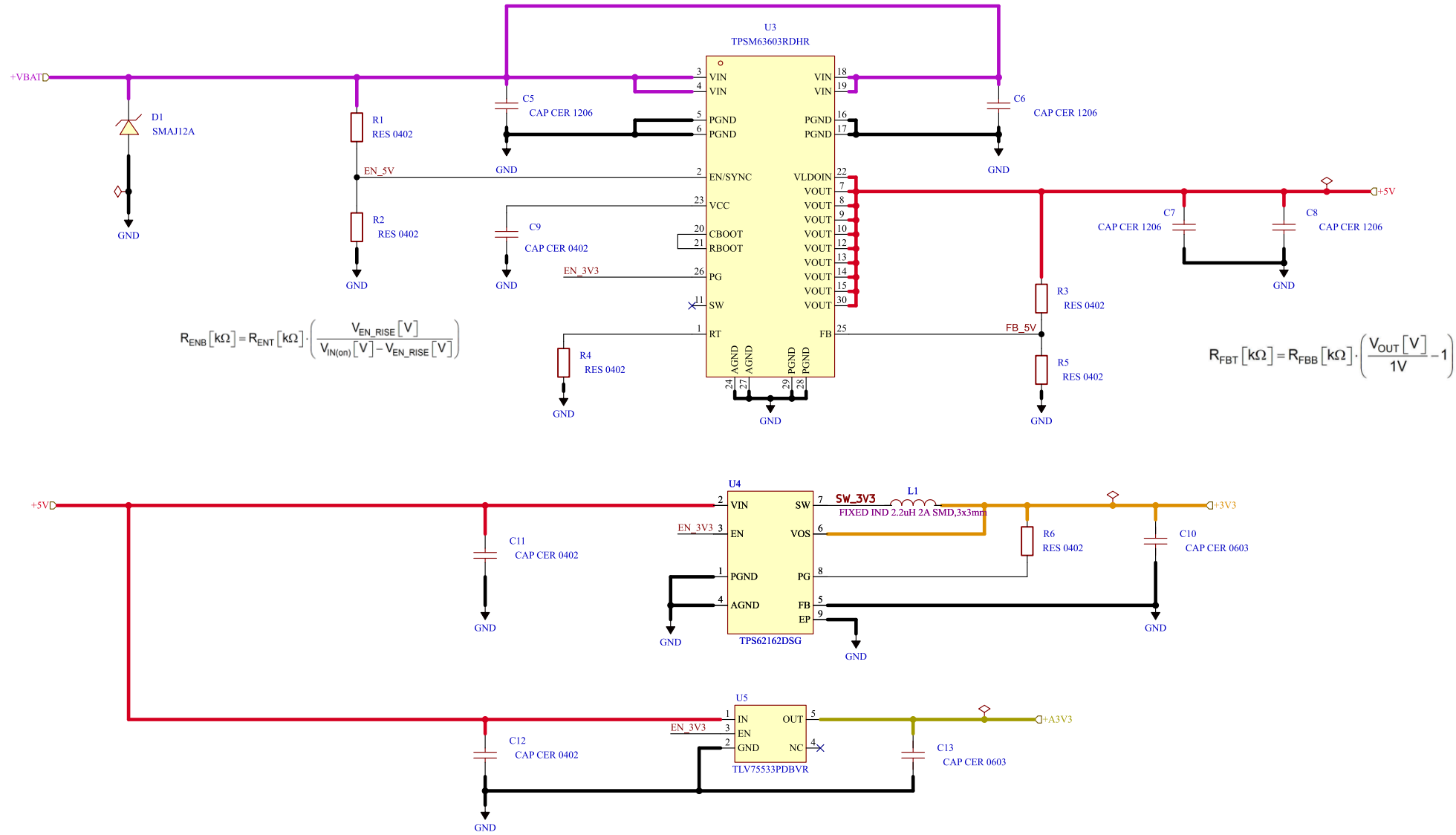


[4] Power - Connectors



Power Connectors - Path Control

[5] Power - Generation

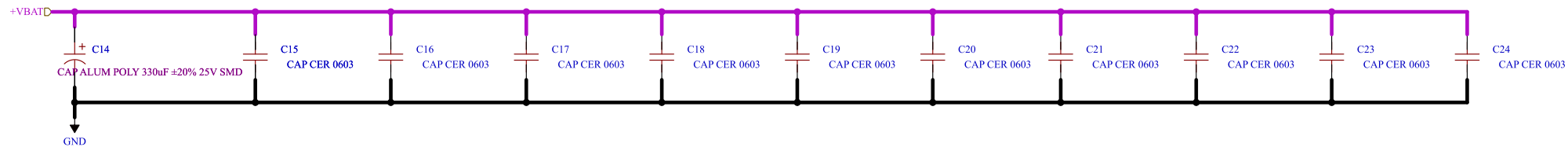


DESIGN NOTE:
PI filter with inductor would give better EMC performance but has higher bulk capacitance requirements. Ferrite is a good compromise.

LAYOUT NOTE:
Route feedback traces away from switching node.

LAYOUT NOTE:
Minimize switching node loop areas.

Voltage Regulators



DC BULK

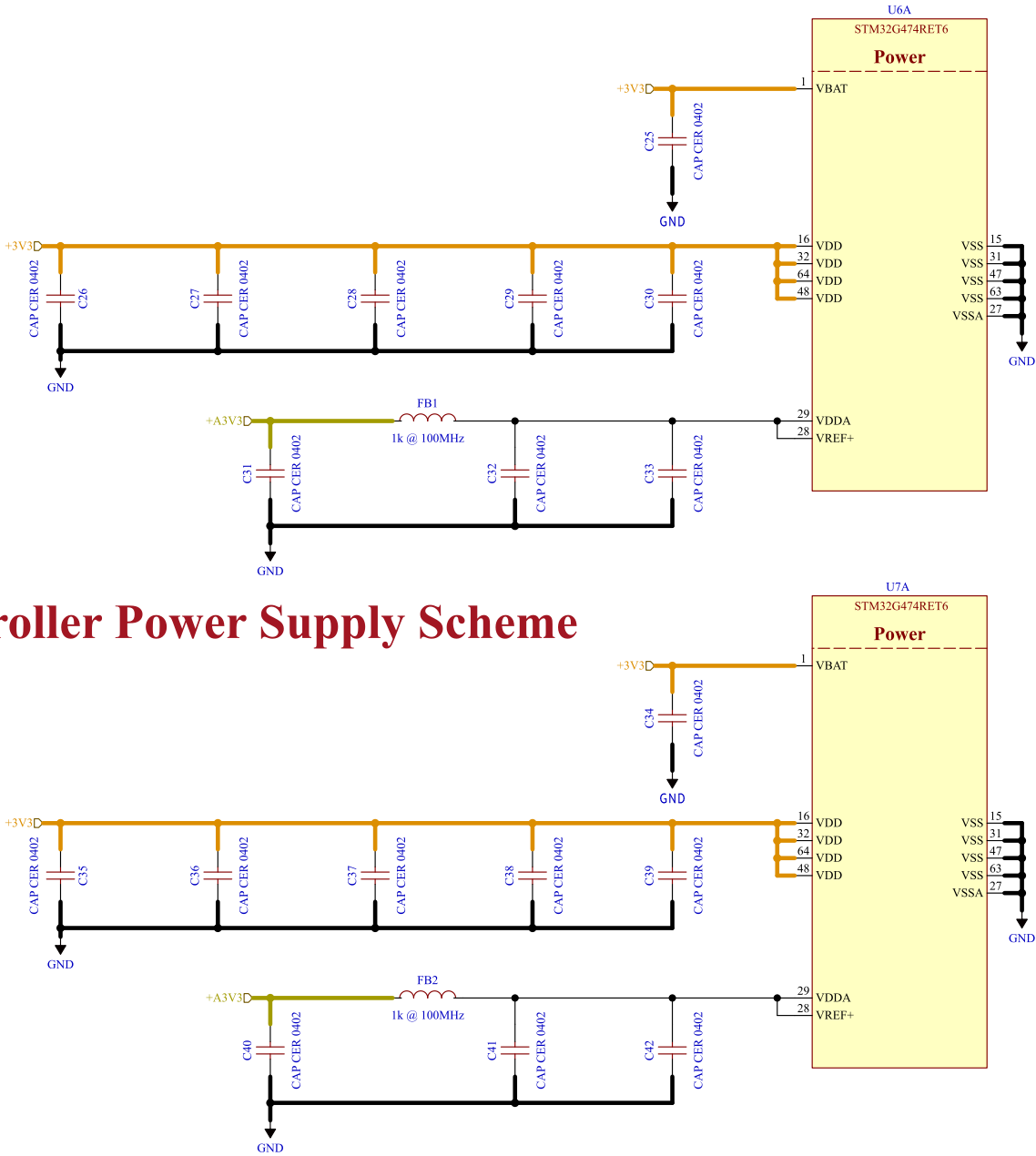
DC-Link Bulk Capacitors. 330uF + 10 x 10 uF = 430uF (35V)

Test Points

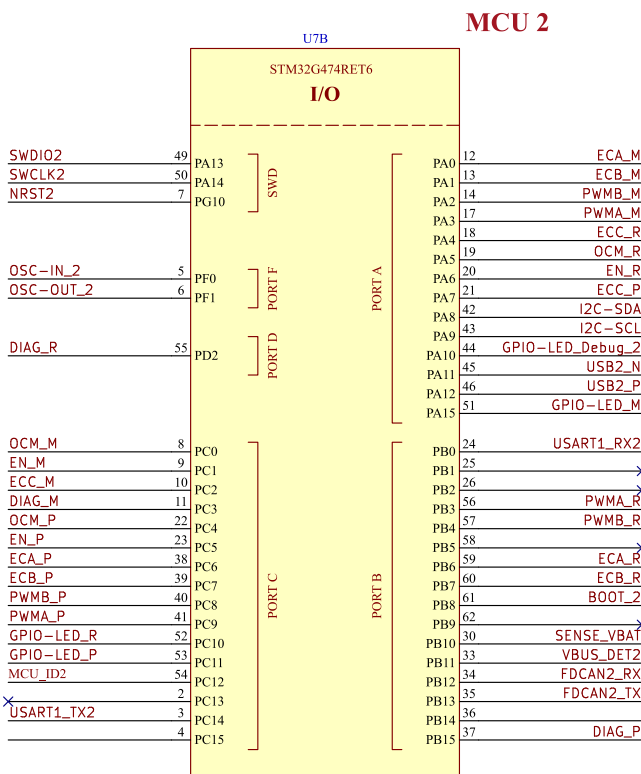
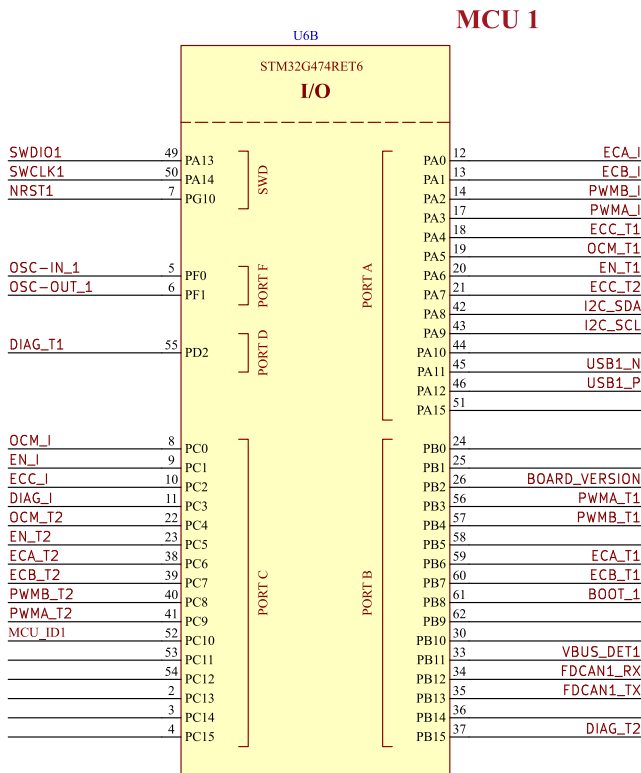


Microcontroller Power Supply Scheme

LAYOUT NOTE:
Place each capacitor close
to the relevant pin.
Minimize loop inductance.



[7] MCU - IOs



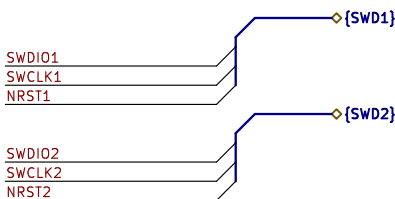
DESIGN NOTE:
We use the LQFP64
version to have extra
pins for controlling an
external fan.

DESIGN NOTE:
Do not forget to flash
option bytes when
flashing for the first
time.

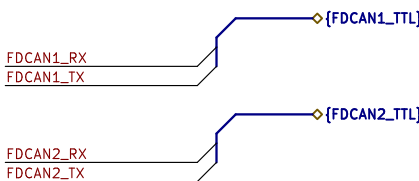
LAYOUT NOTE:
Footprint deviates from
datasheet and IPC7351 to
allow for soldermask sliver
between pads according to
PCBWay design rules.

SENSE_VBAT

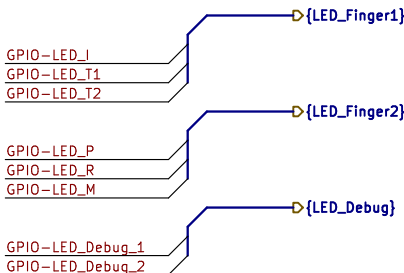
Battery Voltage Sense



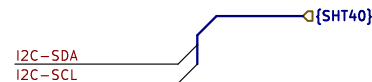
SWD (Debug) Port



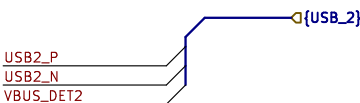
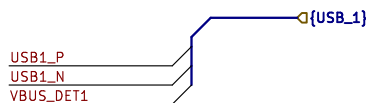
5Mbps FD-CAN Bus



LEDs



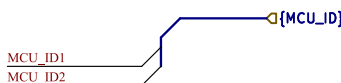
Temperature and Humidity



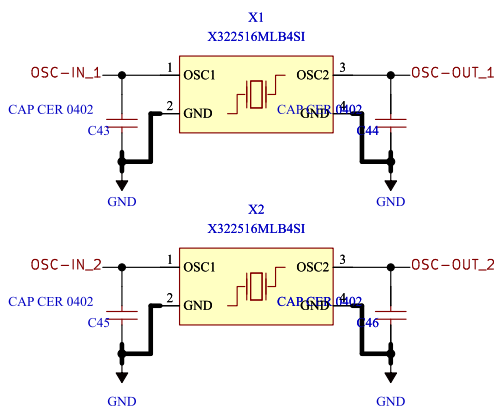
USB PORT

BOARD_VERSION

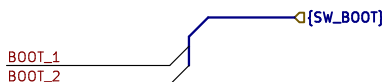
Board Version Selection



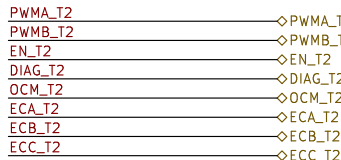
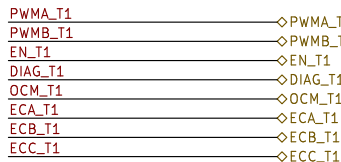
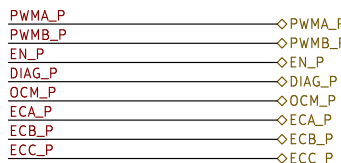
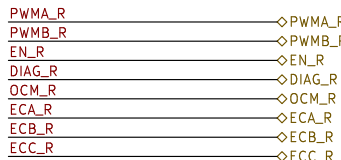
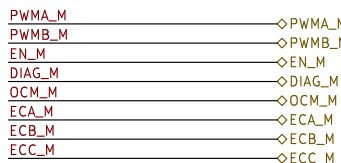
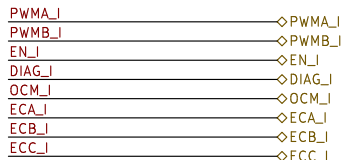
MCU ID



Oscillator HSE



SWITCH



Motor Driver

Battery Voltage Sense

SENSE_VBAT

Board Version Selection

BOARD_VERSION

SWD (Debug) Port

NRST2

SWCLK2

SWDIO2

5Mbps FD-CAN Bus

FDCAN1_TX

FDCAN1_RX

FDCAN2_TX

FDCAN2_RX

LEDs

GPIO-LED_I

GPIO-LED_T1

GPIO-LED_T2

GPIO-LED_P

GPIO-LED_R

GPIO-LED_M

GPIO-LED_Debug_1

GPIO-LED_Debug_2

Temperature and Humidity

I2C-SDA

I2C-SCL

USB PORT

USB1_P

USB1_N

VBUS_DET1

USB2_P

USB2_N

VBUS_DET2

MCB to MB - I2C

I2C_SCL

I2C_SDA

Buzzer

MCU ID

MCU_ID1

MCU_ID2

Test Points

MD Index

PWMA_I

PWMB_I

EN_I

DIAG_I

OCM_I

ECA_I

ECB_I

ECC_I

MD Middle

PWMA_M

PWMB_M

EN_M

DIAG_M

OCM_M

ECA_M

ECB_M

ECC_M

MD Ring

PWMA_R

PWMB_R

EN_R

DIAG_R

OCM_R

ECA_R

ECB_R

ECC_R

MD Pinky

PWMA_P

PWMB_P

EN_P

DIAG_P

OCM_P

ECA_P

ECB_P

ECC_P

MD Thumb 1

PWMA_T1

PWMB_T1

EN_T1

DIAG_T1

OCM_T1

ECA_T1

ECB_T1

ECC_T1

MD Thumb 2

PWMA_T2

PWMB_T2

EN_T2

DIAG_T2

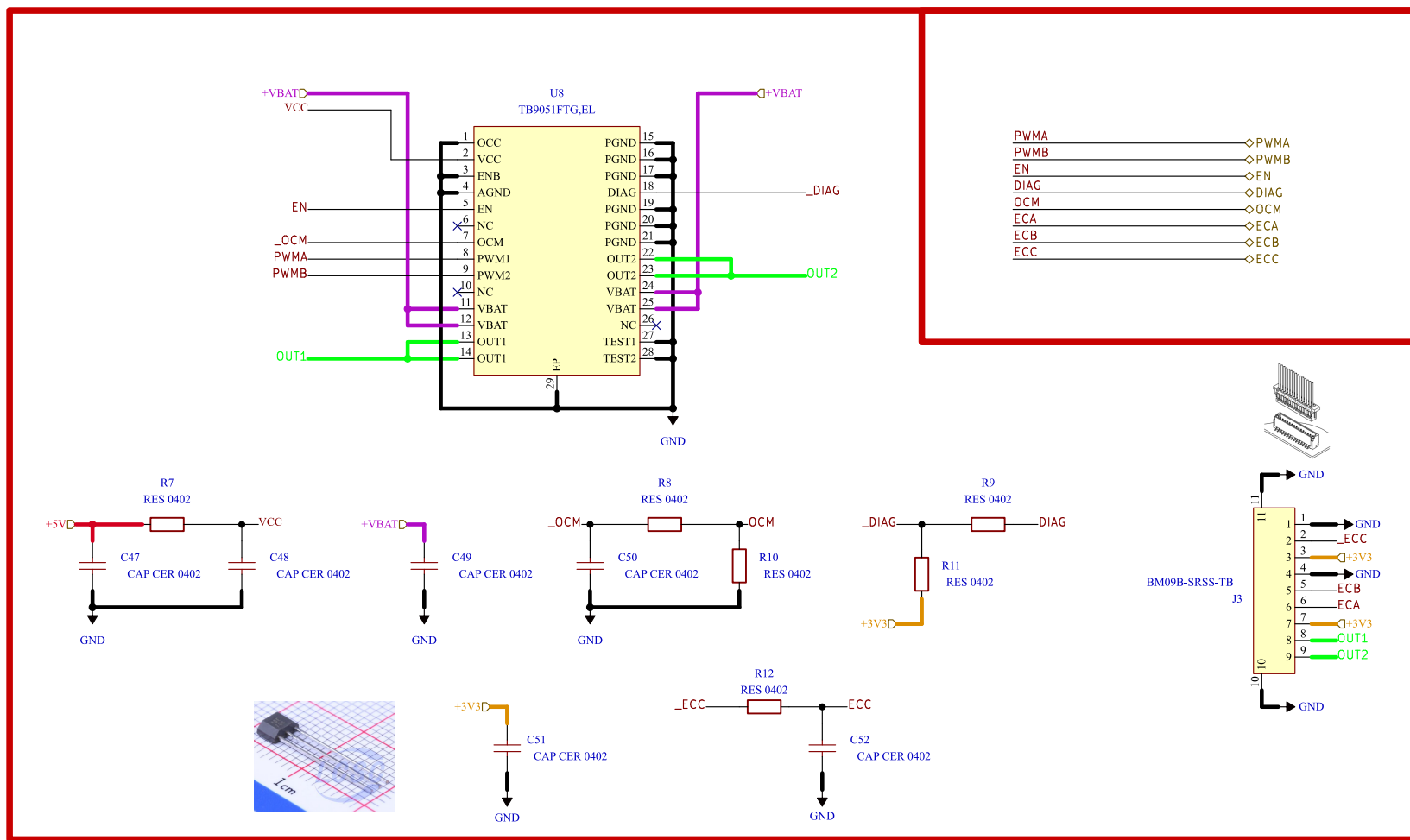
OCM_T2

ECA_T2

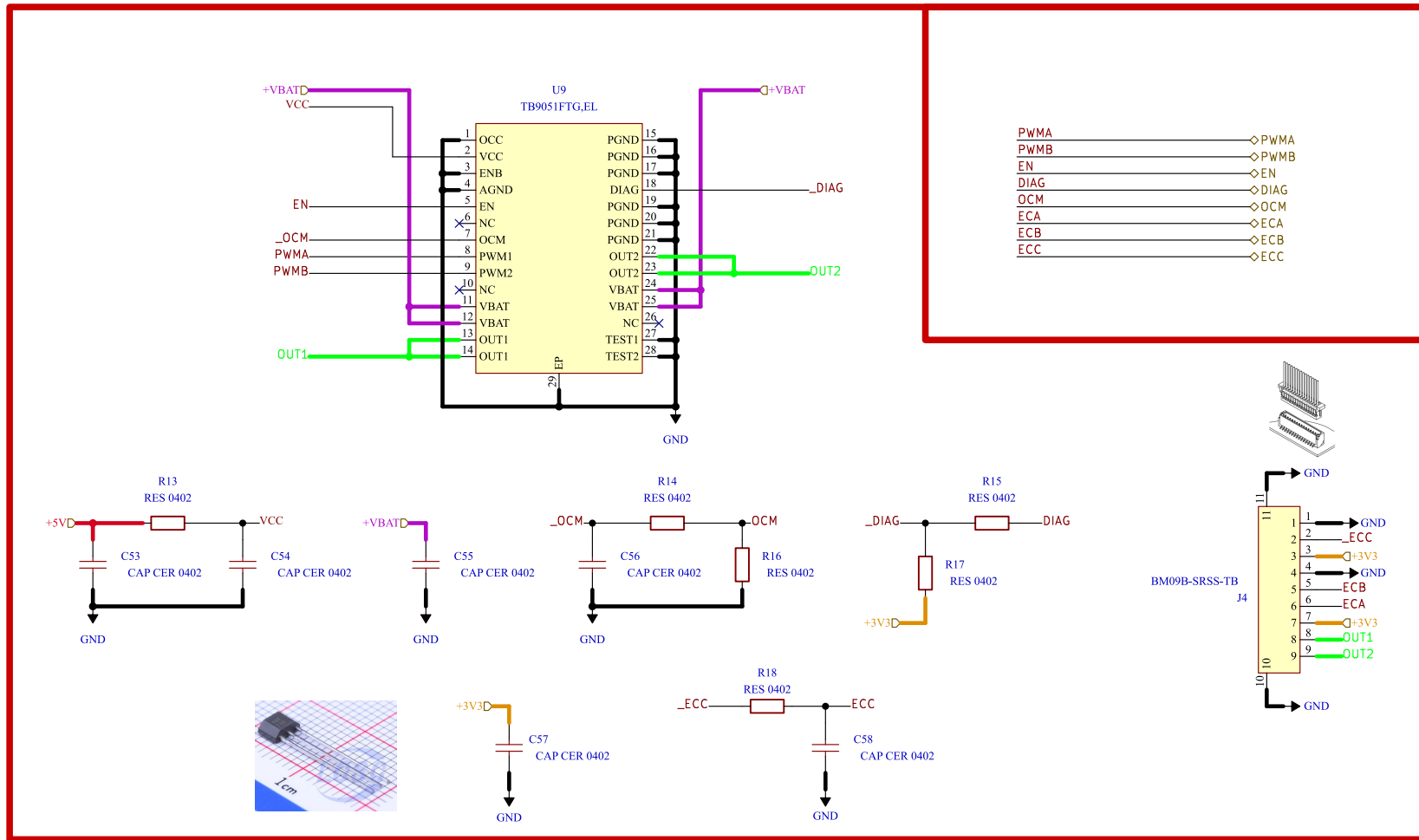
ECB_T2

ECC_T2

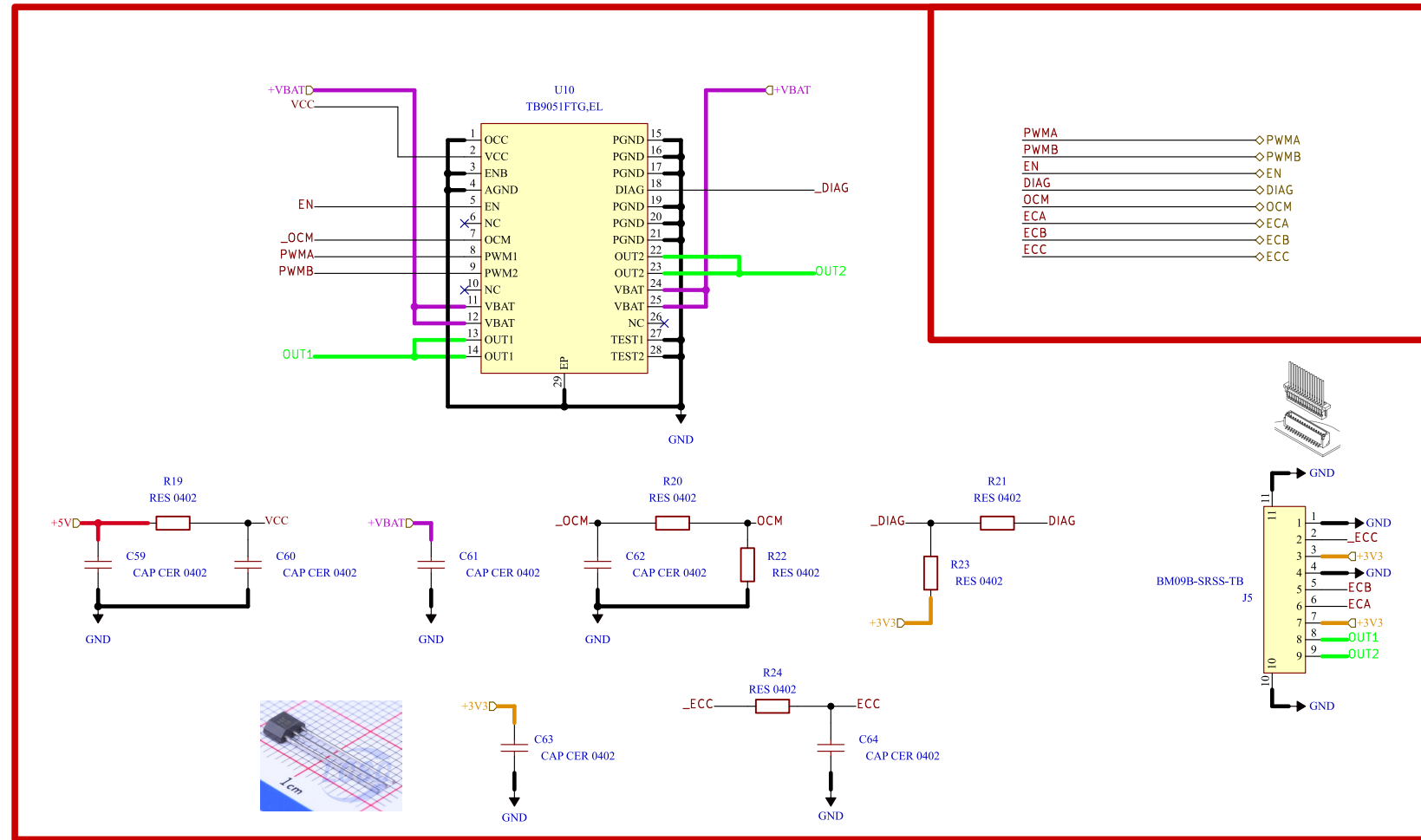
[8] Motor Control 1



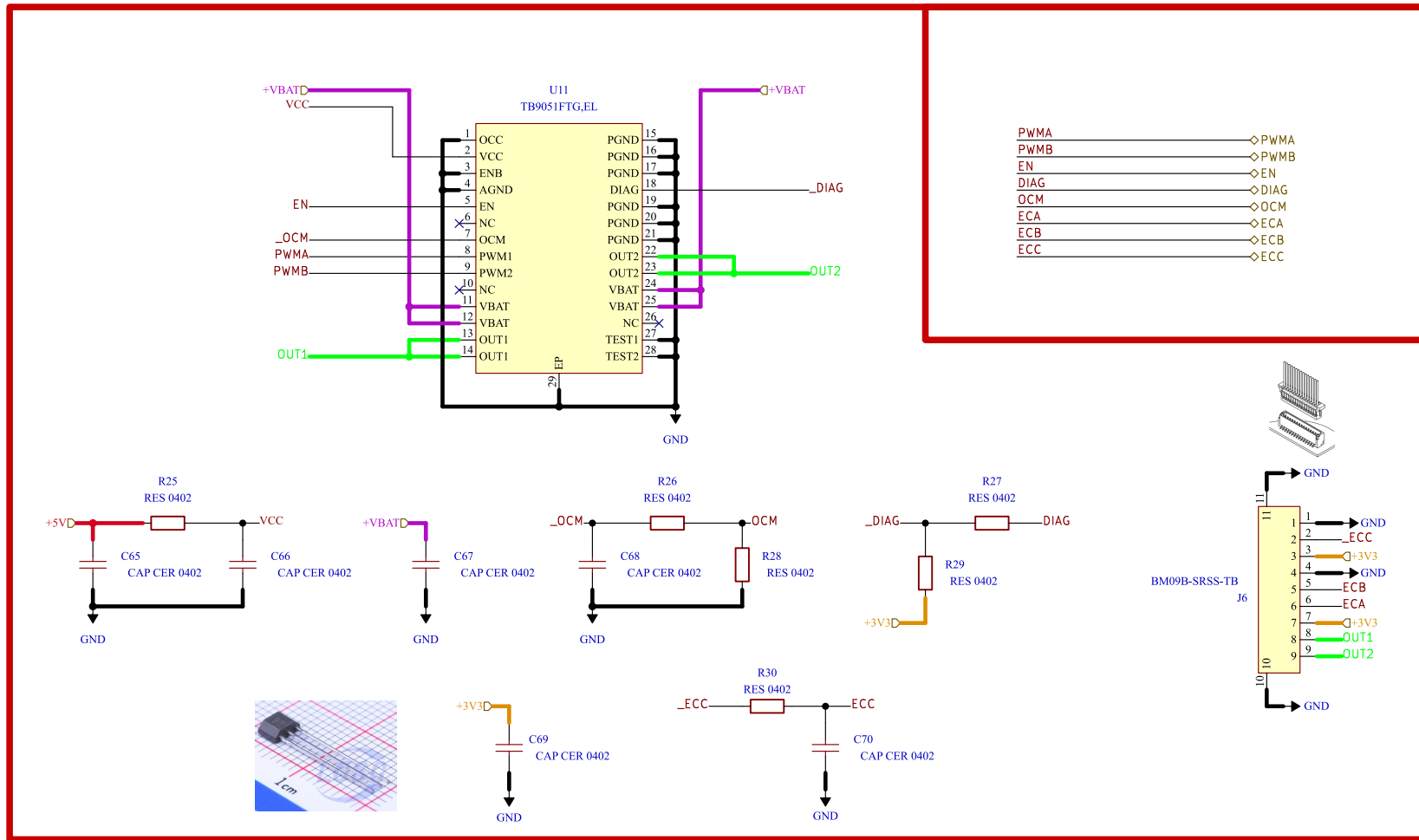
[9] Motor Control 2



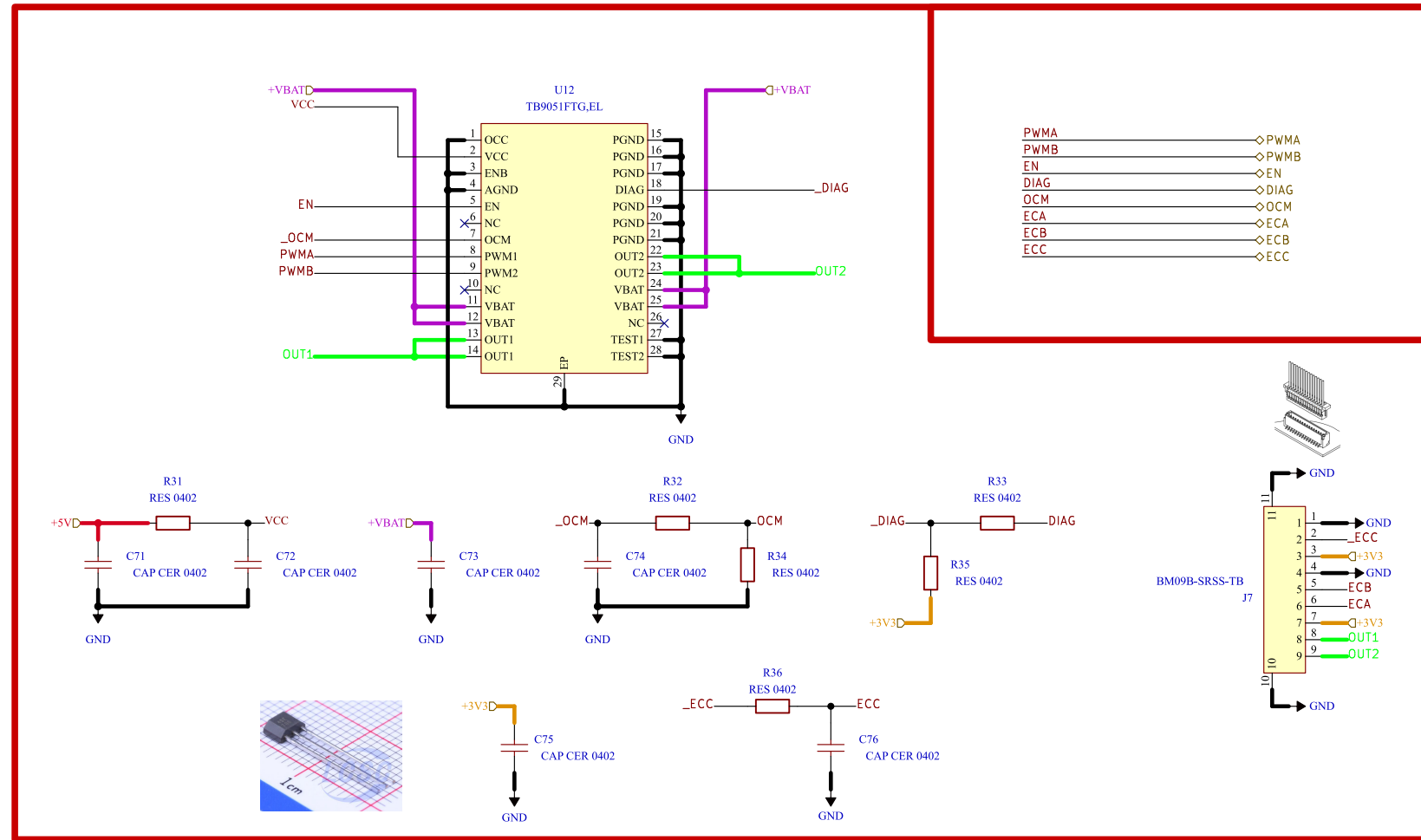
[10] Motor Control 3



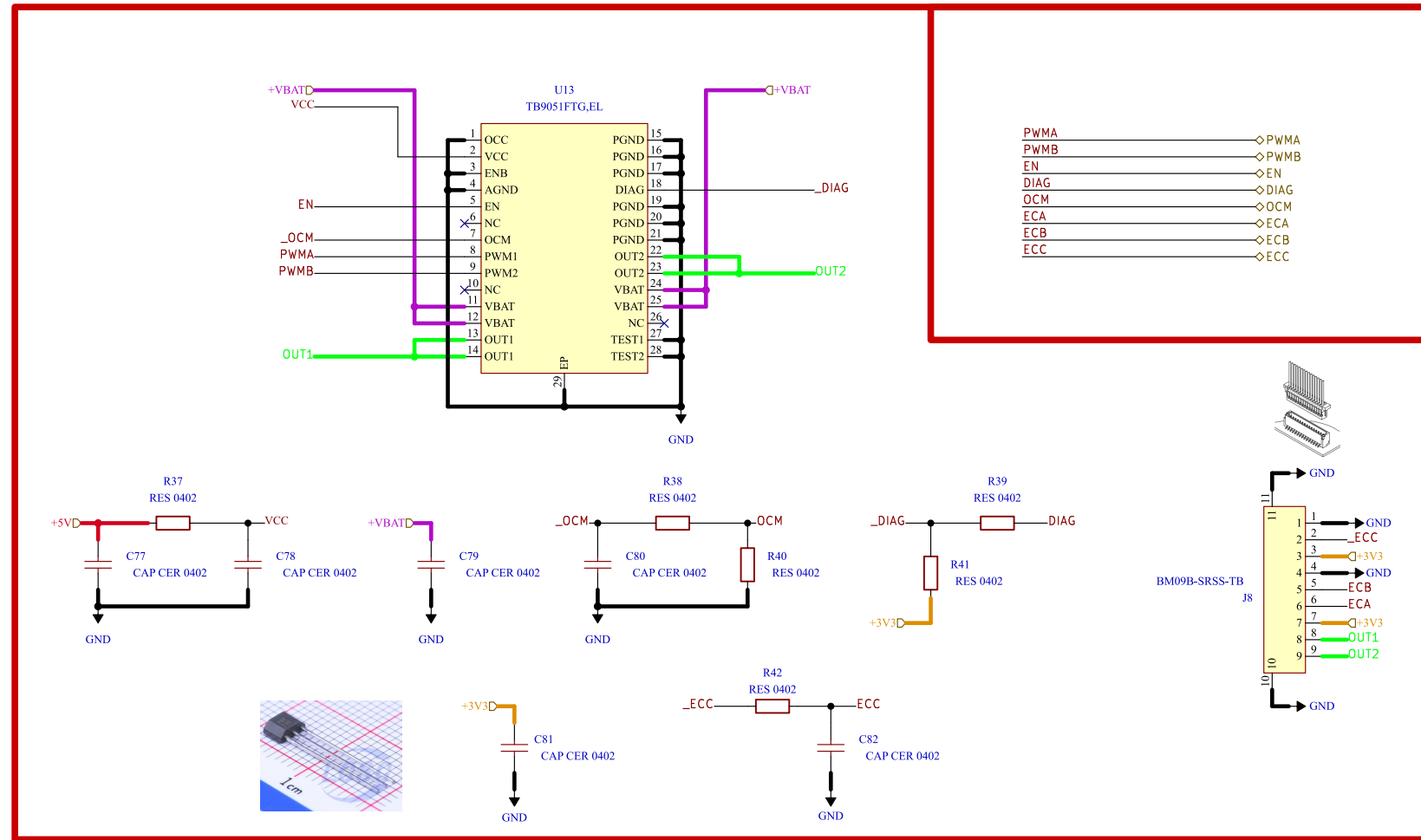
[11] Motor Control 4



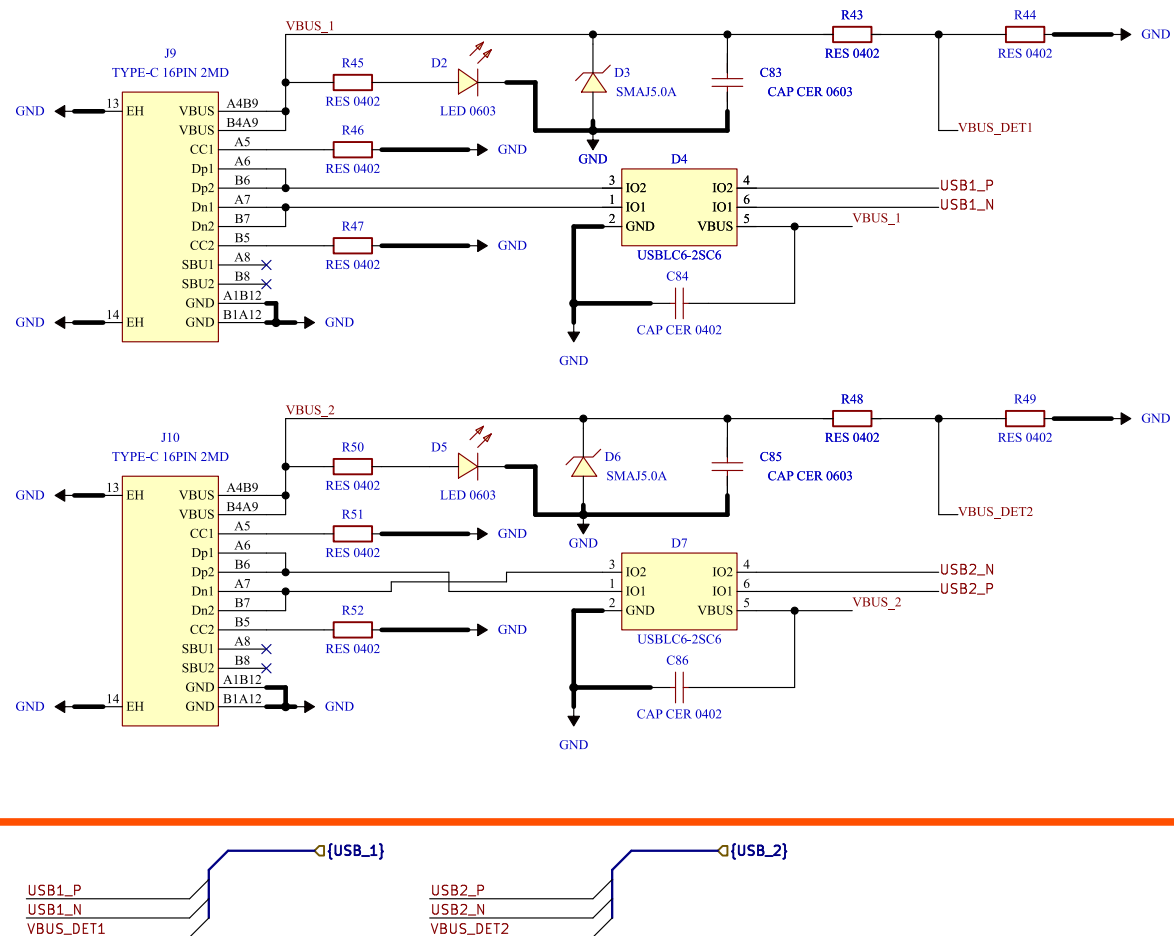
[12] Motor Control 5



[13] Motor Control 6



[14] Interface - USB 2.0



DESIGN NOTE:

[1] VBUS NOT connected to +5V/+3V3. Board is self-powered(LiPo). Sense + LED only. No backfeed to host.

[2] LED direct from VBUS @ 1mA. USB2.0 suspend limit 2.5mA. LED 1mA + divider 33uA = 1.03mA (41%). Do NOT raise to 5mA.

[3] VBUS_DET required - G474 has no dedicated VBUS pin. Self-powered device must not pull up D+ when VBUS=0. FW controls DPPU bit in USB_BCDR. 68k/100k: VBUS 4.40V -> 2.60V > VIH 2.31V. Debounce 100ms.

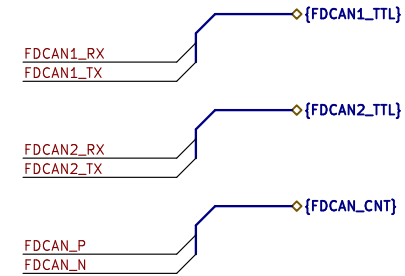
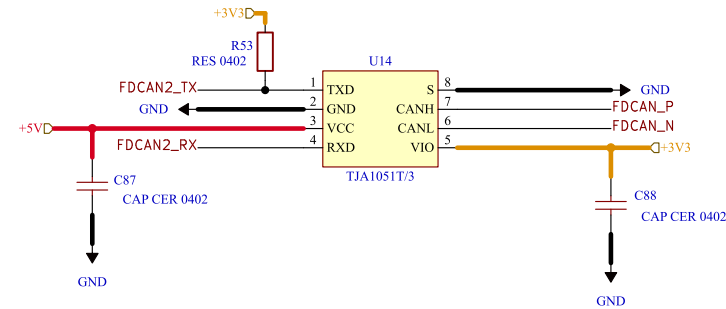
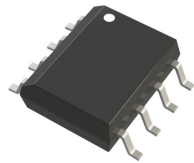
[4] D+/D- direct to PA12/PA11. No 22R series. No external 1k5 pull-up (internal RPUI 900-1500R).

[5] USB clock: PLL-Q 48MHz from HSE 8MHz. ST DFU bootloader uses HS148+CRS - flashable without crystal.

[6] Both USB ports fully isolated - no shared nets.

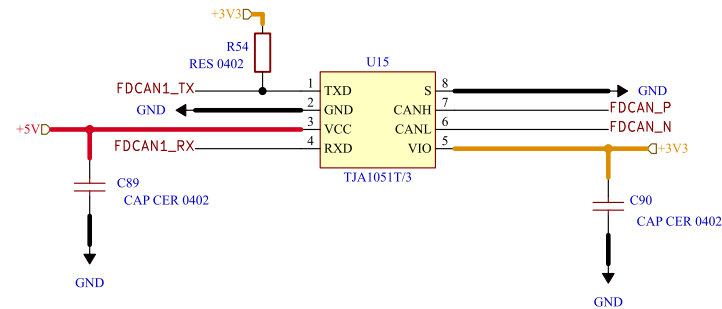
[7] GROUND LOOP WARNING: USB from PC ties host GND to power GND (4.6A peak @20kHz). Use isolator (ADuM3160) or isolated USB-UART when debugging with motors running.

[15] Interface - FD-CAN



DESIGN NOTE:
Silent mode disabled, Pin "S" to ground.

DESIGN NOTE:
120-ohm termination in Interconnects sheet.

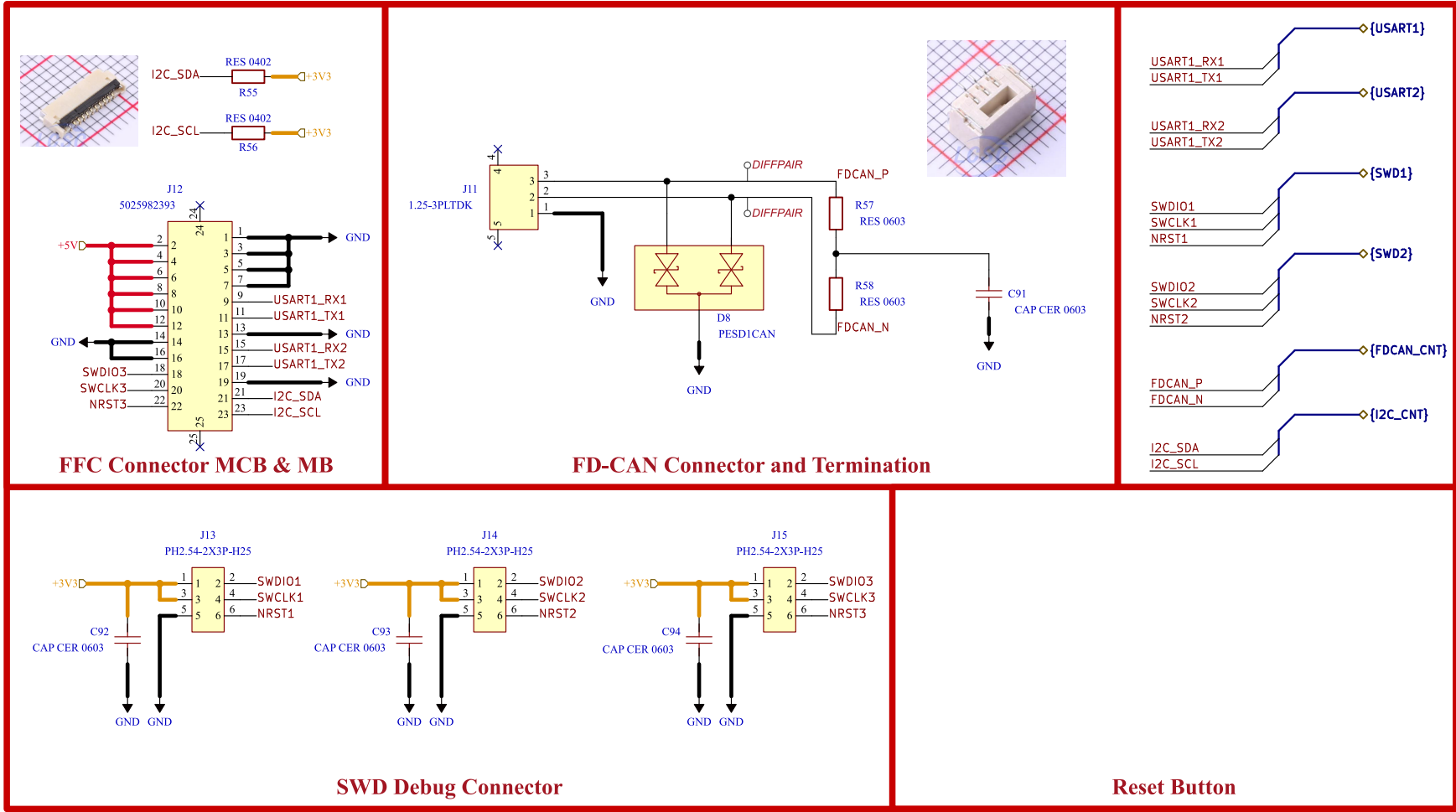


LAYOUT NOTE:
Place close to FD-CAN connector if possible to reduce stub length.

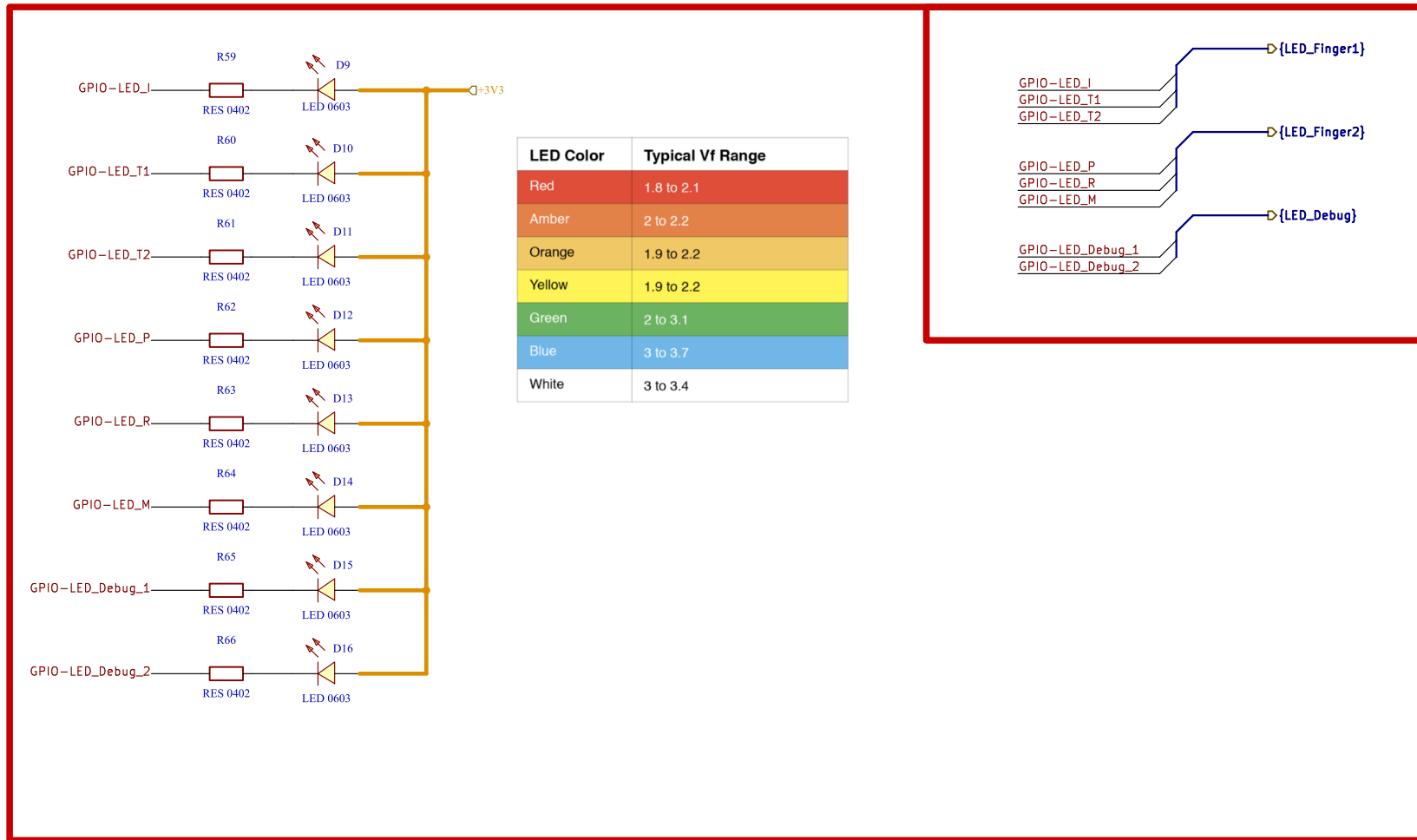
LAYOUT NOTE:
We decided to route it further away to minimize risk of ground lift on the RX/TX lines due to high currents.

Fault Protected FD-CAN Transceiver

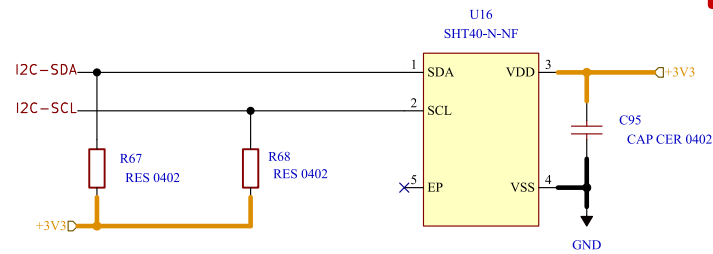
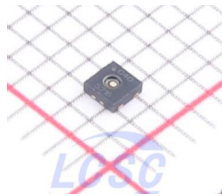
[16] Interface - Interconnects



[17] User - LED Indicators & Buzzer



[18] Sensing - Temperature & Humidity



I2C-SDA
I2C-SCL

{SHT40}

[19] Sensing - Battery

DESIGN NOTE:
Battery voltage sensing divider for 2S LiPo battery.

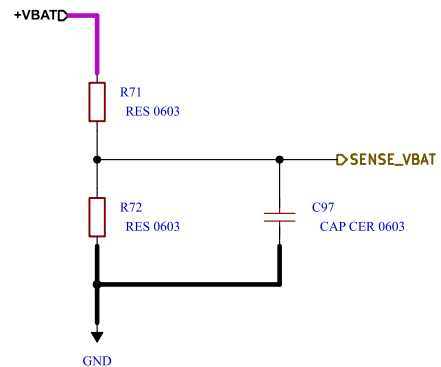
R1 = 100k Ω (Top)
R2 = 47k Ω (Bottom)

Divider Ratio:
 $V_{ADC} = V_{BAT} \times (47 / (100 + 47))$
 $V_{ADC} = V_{BAT} \times 0.3197$

2S LiPo:
Max battery voltage (8.4V): 2.69 V
Nominal voltage (7.4V): 2.37 V
Low battery voltage (6.4V): 2.05 V
Cutoff voltage (6.0V): 1.92 V

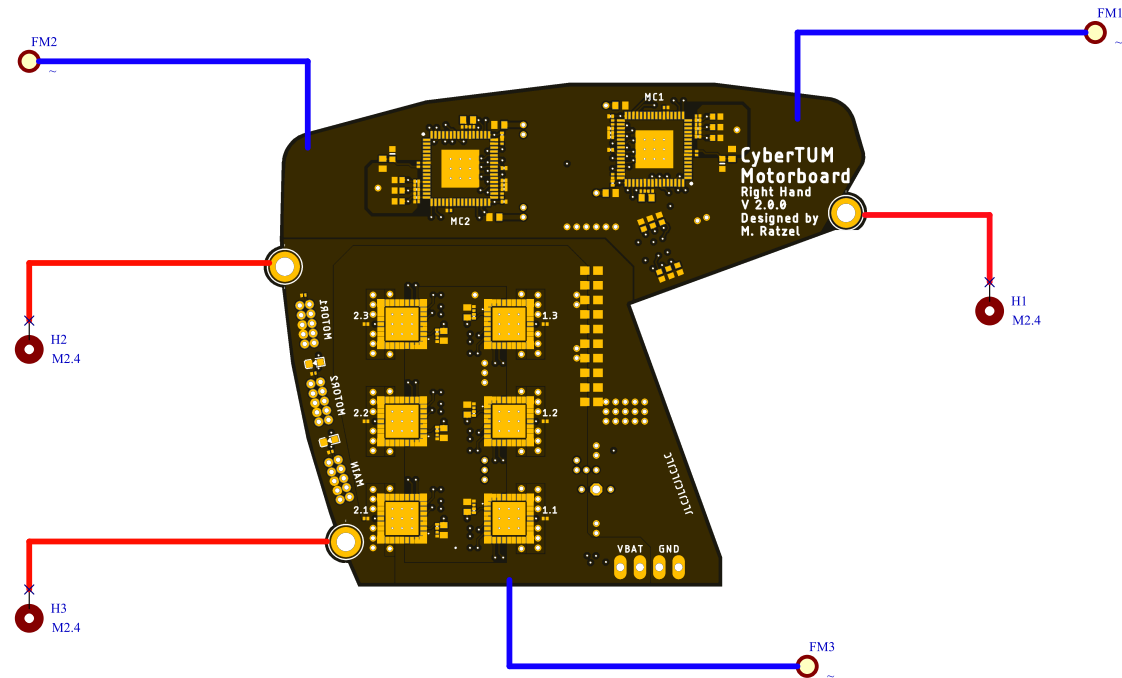
Maximum divider current:
 $I = 8.4V / 147k\Omega = 57 \mu A$

Notes:
- Designed for 3.3V ADC input.
- Add 100nF capacitor at ADC input for noise filtering.
- Use 1% resistors or better.
- Use low temperature coefficient resistors (≤ 100 ppm/ $^{\circ}C$) for improved measurement accuracy.



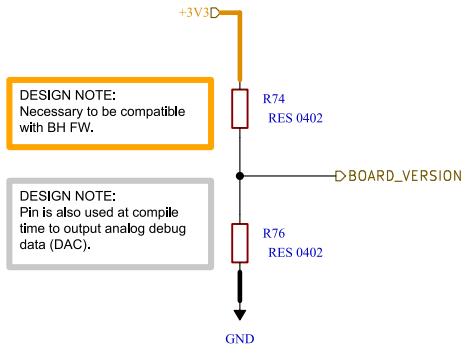
Battery Voltage Sensing

[20] Misc - Holes, Fiducials



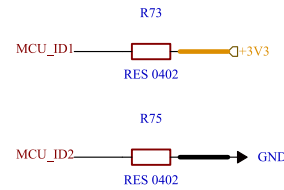
Mounting Holes and Fiducial Markers

[21] Misc - Board Version, Switch

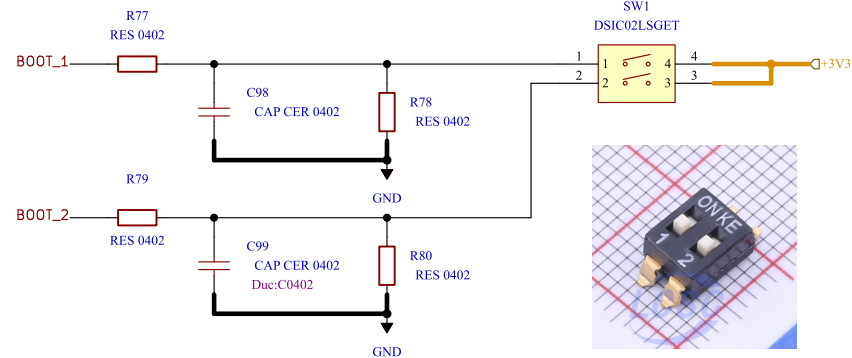
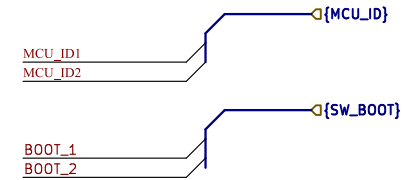


		TOP	BOTTOM
MCB	r1.1, r1.2	×	100k
	r1.3	100k	×
MB	v1.0-1.1	100k	100k

Board Version Resistor Divider



MCU ID



SWITCH BOOT MODE

A

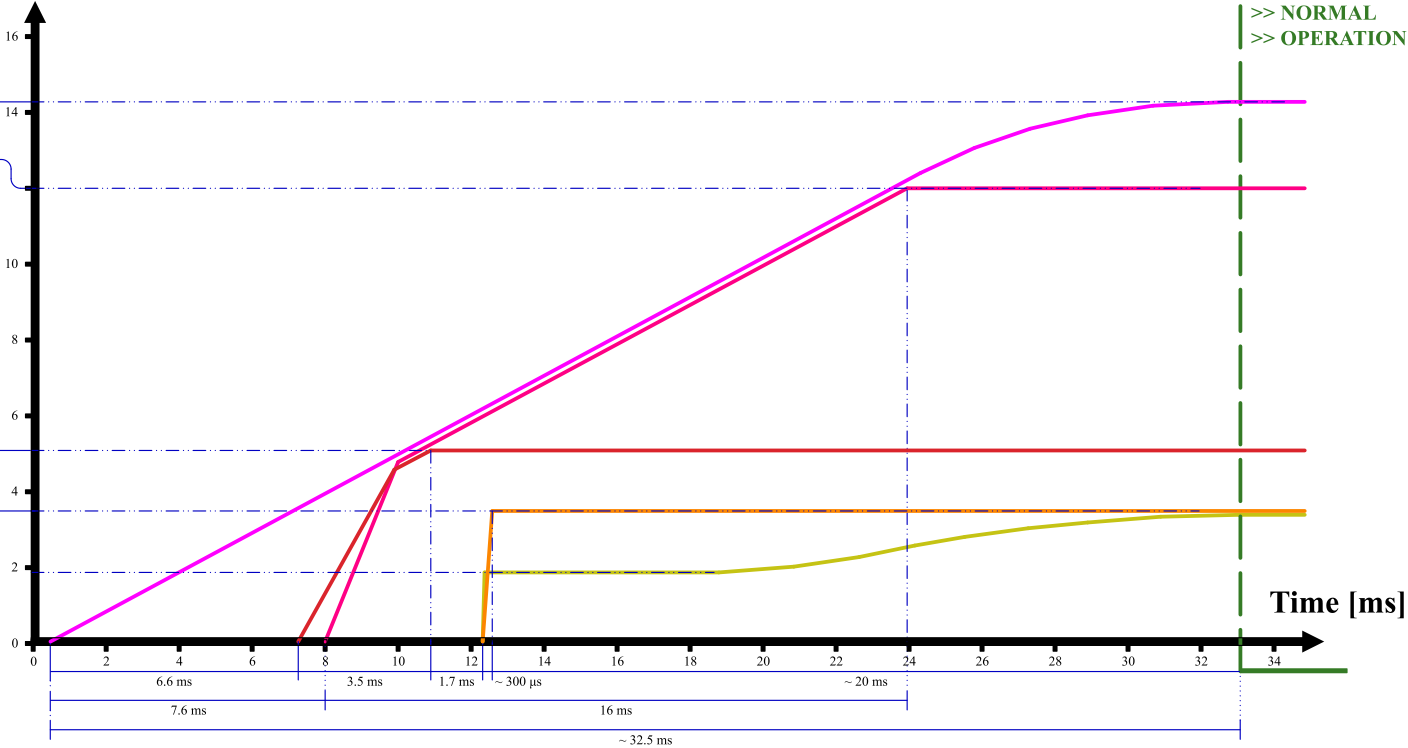
B

C

D

NAME	SOURCE	LEVEL
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%

Voltage [V]



	1	2	3	4	5	6
A						
B						
C						
D						

[23]